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Enhancing Computer Science Education: Gamification for Developing Technical and Transversal Competences

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***Abstract*— Games are created for humanity to get money, for business for the creators and to get fun as an entertainment for clients or gamers, however, they are focused on helping to get pleasant experiences. In recent years, due to the increasing use of technology, miniaturization and portability of electronic devices, new educational strategies for digital natives students, and the proliferation of mental illnesses associated to anxiety stress and depression, games have played an important role in the wellness of humanity and have moved to fields where they were not commonly used. Education is a field that has been visibly invaded by games; therefore, two teams of research fields have been created: games-based learning education and gamification. Game-based learning uses games to enhance the learning experience whilst gamification applies elements of games in non-games contexts. This paper explains the development of a gamification exercise for university students. The gamified proposal was focused of software development projects using technical skills and motivates the participation and engagement of students in basic activities as listening, reading, speaking, writing and develops transversal skills activities as negotiation, taking decisions and testing software. Finally, the purpose of the exercised is to decrease the plagiarism of software projects.**

Keywords— Digital Native Students, Education, Entertainment, Game-Based Learning, Gamification, Serious Games

I. INTRODUCTION

Humans use games usually for entertainment and fun, it is commonly related to other words as play, entertainment and fun, and however, it have recently been related in other fields such as education [1], [2], [3], [4], human resources [5], social skills [6] collaboration [7], business [1], cognitive skills [8], among others. In [9] was proposed a formal definition of game "A game is a system in which players engage in an artificial conflict, defined by rules, that results in a quantifiable outcome". The definition includes four elements: the system, the conflict, the rules, and the outcome. The system refers to the game in general, the conflict can be associated with the goal that is pursued, the rules are the guidelines or that each of the players must follow and the outcomes are the benefits obtained by winning the game.

In [10] was defined "A game is a system in which players engage in an abstract challenge, defined by rules, interactivity, and feedback, that results in a quantifiable outcome often eliciting an emotional reaction". Like [9] [10] includes in the game definition the words system, rules and outcome, but the definition includes other elements as abstract challenge, interactivity, feedback and emotional reaction. In this definition, there are a challenge and an emotional reaction, which can be related to the motivations of the player. Additionally, there are interactivity and feedback, which can be related to social skills.

Students are the raw material of the universities and teaching and learning are fundamental processes in university education. Different factors in teaching produce different effects on learning, among which are motivation, participation, interest, and discipline, among others, obtaining positive results in the students experience and impacting the responsibility and the engagement [11]. Pedagogical approaches suggest that the active participation of students in the process of learning is fundamental to improve the learning progress [11] and [12] even though there are cognitive problems [13]. The use of games and game elements in pedagogical approaches engage and motivate students in the learning process [14].

Neurological studies demonstrate that negative emotions as fear, depression, anxiety, stress and boredom activate the amygdala, which triggers high levels of cortisol and norepinephrine that inhibit the hippocampus activation and new learning. In opposite, Positive emotions activate the hippocampus and inhibit the amygdala [15] actually, some research suggests positive or negative impacts on the brain for the exposition to video games [16].

Two most representative approaches used on education are Game-based learning [17] - [18] and gamification [14], [19] and [20]. The approaches have been used interchangeably; moreover, they refer to two different pedagogical methods. On one hand, for game-based learning, the games are designed to enhance the learning progress of the learners. On the other hand, the gamification applies or integrates principles or game design into non-game context. In this sense, for game-based learning, the game is used in the education, but in gamification, game elements are applied in the education context.

II. LEARNING AND ACTIVE METHODS

Active learning refers to combine several innovative pedagogies and educational strategies into quality learning environments to engage the student to actively participate in transformational learning process. In active learning the student has conscientious about the educational necessities, then they assume the responsibility of answer complex question, search information, make decisions, propose solutions, solve problems, explain ideas, among other. The purpose is to involve the students in a learning scenario to help them to develop skills. Active Learning has been used to recreate learning significative experiences, for instance, to teaching nursing graduates [21] and to introduce new concepts for computer sciences students by using a game [22]. An approach to identify several active learning processes is documented in [23].

III. GAME-BASED LEARNING

The use of games in an academic context has pro and cons, then it is easy to find positive and negative perspectives related to the use of games in the classroom. In one hand, there are a concern about the addition that it can cause, the age of the students and the expectation of the experience. On the other hand, the game seeks for increase the motivation and participation of students and allow the students to represents artificial roles, solving problems, interact with others and share information or goals. The game plays an important role in education, especially in child formation, they are based on play games, the difference is that games are not necessarily digitals. In a generic way, a game is an activity digital or not, that can be used to teach, educate, train or reinforce the knowledge of the students at the same time that enjoy the experience [24]. A game can be used as a creative and innovative strategy to teach or train a student or gamer in specific skills that can be technical or transversals whilst get fun. A division of education games into serious games and gamification is proposed in [18], in both cases there have not an entertainment purpose. The authors defined serios games as games used for students for training and practicing. In contrast, gamification use games-like features to help the students. The game Trouble Hunter for computer sciences studies [22]. A game calls the attention of students as try to maintain the motivation, interest and engagements; however, the problems is to maintain the attention and focus it in the learning process, it is not only play it should help the students to improve the skills, take advantage of the engagement, motivation, adaptability, creativity, among others [17].

IV. GAMIFIFICATION

Gamification word was introduced in 2008 and it was widespread in 2010, there are several meanings, and some of them are "the application or game elements in non-game context" [25], also it is defined as the utilization of pedagogical system based on games implementation but within a non-game context. Although gamification can be used in many contexts, the areas in which this term has been used the most are related in other fields such as education [1] [2], teaching [3], learning [1] [4], training [3], human resources [5], social skills [6], [4] collaboration [7], cooperation [4], business [1], marketing strategies [7], health [21] among others.

The systematic review of the literature by [19] produced a list of the most important principles that must have a game for education, then a brief explanation of each one:

Goals: refers to the objectives or the purpose that must be achieved within the game, for example, collecting objects, building cities, guessing words, following a path, finding a treasure, among others.
Challenges and missions: to reach the goals, the players are presented with challenges or missions that they must complete, for example completing a track or leaving a labyrinth.
Customization: To improve the user experience, you must allow the game to be customized and adapted to the users, in this way beginner players are differentiated by difficulty.
Progress: At all times the progress of the players must be visualized. For some games it is necessary to know the overall progress and progress relative to missions, tasks or levels.
Feedback: When the player makes a mistake or loses a challenge, the game should immediately give a comment, in the same way when he or she wins, he or she should immediately receive a prize or reward.
Social engagement: Collaboration or competition should be encouraged; this helps to construct relationships with others.
Accrual grading: Accumulative assessments are better than general assessments. It is better to earn a few points for small tasks constantly, than to earn many points for completing a mission. It is a way to offer a constant stimulus.
Visible status: The visibility of the best players allows them to gain reputation and it stimulates their participation and the competition from others. It is like a table of honor.
Access / unlock content: new features or new content may be visible depending on the points accumulated, for instance, new labyrinths, missions, questionnaires, among others.
Freedom of Choice: There must be several ways to explore and several paths that allow them to complete the goal and accumulate points, for instance complete the missions in different order.
Freedom to fail: The possibility of having several opportunities or attempts to complete activities must be included for all players, even though when they are not in the table of honor or when they are in the last place.
Storytelling: The story should be used to make sense of the game and to motivate players to continue playing; it should be clear, easy to understand and as simple as possible.
New identities: give the possibility of having new identities or roles that allow them to identify themselves within the story. Connect the identities with the game.

Onboarding: Start with simple tasks to introduce the players to the game. Increase the level of difficulty in each level of the game to ensure motivation.
Time restriction: Check carefully the time restrictions that are used to complete the activities.

The list of defined elements is not necessarily included in all the games, that is, the creators or designators define what elements are included in each given game because the purpose can be affected. In fact, [19] explains that there are some that are more commonly used, but that their importance depends on the objectives and purposes of the game. Making a literature review and implementation, [19] highlights the importance according to the implementation or use in the games. According to the classification given in order of importance the following elements are applied: visible state, social commitment, freedom of choice, freedom of failure, fast feedback, goals / challenges, narratives / new identities, content unlock, incorporation, time restriction and personalization. Similarly, [19] and [20] identified eight mechanisms used for games in educational context, in the next section there are an explanation focused on the learning process:

Points: In an academic environment can be considered grades. They can be applied, for example, for the result of a questionnaire or for the progress of a project or document. The points are given for completing the activities and if they all complete them all must receive the points.
Levels / Stages: The levels indicate progress, increase of the skills and abilities; therefore, as the level increases, the complexity of the tasks must be increased. For instance, for a first level the basic concepts can be understood, and for the second and third levels a summary and a model or representation graphically can be done, respectively.
Badges: It is usually called marks of appreciation; they are given to players who stand out. It is a way to increase commitment and motivation. The badge is assigned to the outstanding players, this motivates the competition. For a child, a badge is a happy face or a flag.
Classification tables: In this tables only the most prominent users are shown, you can make several, for example one to highlight those who have gone further, those who have completed more content or those who have obtained higher scores, the who have completed the activities in less time, among others.
Awards and Rewards: It should not be confused with points or badges. They can be small rewards that stimulate participation, such as allowing you to change the clothes of avatar or allow free time or free reading time.
Progress bars: These elements help the user to have a global vision of their progress, which can be measured, for example by the content developed, the successful challenges. Progress is commonly associated to the content.
Storyline: The narrative helps players stay motivated. A structure must be created that allows the player to know the details of the story while playing. The narrative goes hand in hand with the game and must create a context for learning.
Feedback: the feedback allows us to assess the successes and failures and know what can be improved. When the feedback is positive, the player gets points, badges, awards or rewards; in contrast, when it is negative, the player must be allowed to repeat the process. In both cases, the feedback must be immediate, clear and precise. In [20] was proposed other classification that includes and extends the previous ones.

V. Gamification Exercise

The gamification exercise was applied to the Techniques and Practices of Programming course with 30 second and third-semester Computer Science students at a private university in Cali, Colombia. The students, aged between 16 and 24 years, participated in a project that simulated small software companies. These "companies" were formed by teams of three members, and their objective was to develop software and offer software testing services. The exercise lasted 9 weeks and followed the rules below and the next part are the rules of the exercise:

Team Formation
- Each company was composed of three students who chose their teammates.
- Each team had to define a unique software project, different from those of other teams.
- Teams remain intact for the whole-time exercise.

Identification and Initial Capital
- Each company had to have a name to identify their project and team members.
- Each company started with an initial capital of $100.

Company Activities
- The companies engaged in both software creation (internal) and software testing services (internal and external).
- They had to create documentation and write the source code necessary to complete the project.
- Each week, they had to test a different source code or application from another company.

Financial Management
- Money was used to pay workers' salaries and for testing services from other companies.
- Companies earned money by finding errors in other companies' code.
- Payments for salaries and services were made weekly.
- Companies earned points for completing proposed activities and money for errors found in the testing service performed for others.

Roles and Rotation
- Each team had three roles: Project Manager, Development Engineer, and Testing Engineer.
- Roles were rotated weekly so that each member performed each role three times during the exercise.

Submission of Progress
- Weekly, they had to submit the development progress (implementation, source code) using the correct documentation artifacts.

- Errors, including orthographic, semantic, syntactic, compilation, and execution errors, had a price.

Evaluation and Penalties

- The evaluated company could negotiate or try to reduce the requested price for the testing service.
- Points and prizes were awarded to companies with the most profits, the least money loss, the most lines of code, and other achievements.

Exercise Dynamics

In addition to development, companies also offered testing services. Each week, they had to submit an evaluation report for another company and receive one in return. The errors found had a price (see Table 1):

Table 1. Errors and its prices

Error	Price
Misspellings	$0.5
Data entry or data type errors, lack of documentation	$1
Compilation errors, runtime errors, and infinite loops	$2
Code that does not compile	$5
Failure to submit the document or test results on time	$10

Incentives and Penalties

- Students earned money through weekly salaries, which is decided by the company members.
- Teams could apply internal sanctions, such as withholding pay from members who did not participate or delivered incomplete work.
- The professor charged for evaluations or tests with a minimum cost of $2 and a maximum cost of $20, depending on the evaluation.
- Each week, new content and challenges were introduced for the students, affecting the project progress and feedback.

At the end of the exercise, each company had to present their software product in a simulated sales room environment. The company with the highest sum of money at the end of the exercise received additional points in the final evaluation.

This exercise aimed not only to teach technical skills but also to foster teamwork, project management, and financial decision-making, creating a comprehensive and motivating educational experience.

VI. Results

The most important result is that there was zero plagiarism, in the past 4 of 15 teams used plagiarism. All projects were completely built and developed by the students as homework. In the classroom, the progress of each member of the company was evaluated, doubts were solved, and suggestions were made by the professor and the classmates. However, it can be said that in this experiment, 7 of the 10 teams got involved, submitted the work on time, tried not to lose money, tried to gain money with the tests, they made the tests more exhaustive trying to earn more money. It is also evident that they began to use strategies that allowed them to earn more money and lose less, improved the negotiation of the tests, increased the technical vocabulary, tried to advance faster in the development of source code to earn more points and fulfill all that they were asked, among others.

On several occasions students asked project questions outside the class, by email and consulted other teachers, in relation to past courses, students were more active in the consultations. Additionally, they began to arrive early and to meet with other classmates to talk about tests, also during the class and at the end, repeatedly made comments on the matter.

In relation to the same course in the last semester, these students know that they are less dispersed with the cellphone, and on very few occasions they are seen playing during the class. Only three of the ten companies present problems. Two of the three companies have a student who from the beginning has not shown any interest in class work, despite the insistent observations, they do not do anything of the individual, and they enter and leave the classroom frequently. Two companies have a student who expressed their desire to change careers in the next semester. A company has the two mentioned problems, a disinterested student and a student who wants to change of career. The company with only one interested member stopped working in week 6. The other two teams advance, but they are stuck, they have faults, they run out of money, and they have problems. At the end four students needed to repeat the course, three of the four belonged to the team that stopped in week 6. The students were permanently consulted to see if they were bored and only one in 30 felt that the game was boring. The students were easily involved in activities of reading new material, writing reports, negotiating user tests, making presentations, revisions of source code, tests, among others. Table 2 provides a concise summary of the gamification exercise outcomes, showcasing successes and areas of challenge encountered during the implementation.

VII. Discusion

The gamification initiative aimed to immerse students deeply in learning activities relevant to computer science, integrating both curricular and extracurricular elements. While students engaged in programming and testing tasks intrinsic to their curriculum, they also tackled software analysis and design—skills typically underemphasized in their coursework.

This holistic approach was pivotal in fostering connections between disparate facets of knowledge, thereby enhancing comprehension and application.

In addition to technical competencies, the exercise emphasized the cultivation of transversal skills such as negotiation, communication, leadership, and collaborative problem-solving. Unlike traditional educational settings where students often work in isolation on individual projects, this exercise required them to assume roles like project managers, testing engineers, and software developers. This role-playing not only exposed students to diverse responsibilities but also broadened their vocabulary and proficiency in various domains.

Furthermore, collaborating in different roles facilitated stronger interpersonal connections among team members, enabling them to appreciate diverse perspectives and approaches.

Table 2. Rules of the Exercise

Metric	Result
Instances of plagiarism	0 (compared to 4 out of 15 teams previously)
Total teams	10
Teams fully engaged and submitting on time	7
Teams struggling	3
Projects developed and completed	All projects
Students who needed to repeat the course	4
Teams with no issues	7
Students disinterested from the beginning	2 (in separate teams)
Students wanting to change majors	2 (in separate teams)
Teams stopping work before the end	1 (stopped in week 6)
Students finding the game boring	1 out of 30
Increased engagement (e.g., early arrivals, consultations)	Observed
Distraction by cellphones	Significantly reduced
Active participation in reading, writing, negotiating, presenting, and testing	High

The financial dimension of the exercise provided valuable insights into fiscal management within simulated corporate environments. Students learned to optimize company expenditures and maximize revenues through strategic management of testing services. Interestingly, negotiations surrounding testing services prioritized social skills over mere financial gains, underscoring the importance of effective interpersonal interactions in business transactions.

Teams operated synergistically, not only presenting their projects cohesively but also conducting rigorous evaluations of peers' software. This collaborative effort not only elevated project quality but also deepened students' understanding of software evaluation methodologies and the criticality of rigorous quality assurance practices.

Ultimately, the gamification exercise effectively facilitated the holistic development of students' technical and transversal skills through an active learning approach grounded in gamification principles. By bridging theoretical knowledge with practical application, fostering collaboration, and emphasizing interpersonal skills, this pedagogical strategy enriched the learning experience and prepared students comprehensively for the multifaceted challenges in their future careers in computer science.

VIII. Conclusions

The development of the game allowed students to improve their skills, include content that is not easily included, such as the development of tests, work with simulated companies, representing roles, developing software tests negotiation and making decisions. It was also possible to recreate systems such as salary, and strengthen responsibility, collaboration, cooperation and competence. Finally, it was possible to develop software projects while plagiarism was reduced to zero.

The management of the project was done through excel and the Moodle learning management system, however, in the future it is planned to use a project management tool that allows collaboration, cooperation, interaction between the students and documentation of the process. In addition, it is expected that the results can be visualized in a funnier way.

The commitment of the students greatly favored the development of the game, in terms of content, it could be completed and reinforced with necessary topics, such as tests, business strategies. It is recommended to guarantee weekly prizes, points or grades for weekly activities, to constantly evaluate the progress of the project and to socialize the projects constantly. It is highly suggested to reduce the number of deliveries from 9 to 6, to avoid deliveries on exam dates as much as possible, and to extend the work period from one week to two weeks, to have a more significant advance.

IX. Future Work

While this exercise provides valuable insights into the effects of gamification on computer science students and suggests strategies to increase active student participation, incorporating extracurricular competencies, future research could explore more comprehensive scenarios. One avenue is to expand the study to include multiple courses and professors, forming larger teams and extending the duration. This would enable a deeper understanding of how these activities impact student learning across various domains, potentially adapting the exercise for more complex courses.

Moreover, incorporating advanced computational techniques such as artificial intelligence (AI) and machine learning (ML) could enhance the effectiveness. AI and ML algorithms could be employed to analyze student performance data, optimize team dynamics, and personalize learning experiences based on individual progress and preferences. Integrating qualitative research methods, such as focus groups and in-depth interviews, alongside these computational approaches could provide nuanced insights into students' subjective experiences and perspectives.

Finally, considering the influence of cultural and contextual factors on the student-professor relationship could further refine the design of collaborative projects, potentially organizing them as modular software companies within a unified class framework. This approach could foster interdisciplinary collaboration and prepare students for real-world challenges in the evolving field of computer science.

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